Multilingual Financial Fraud and Scam Detection in India's UPI and SMS Ecosystem

Abstract

This project presents a multilingual system designed to detect financial fraud and scam calls within India's rapidly expanding UPI and SMS ecosystem. With the widespread adoption of digital payment platforms, users are increasingly exposed to fraudulent activities carried out through voice calls and text messages, often exploiting trust and lack of awareness. To address this issue, the proposed system leverages advanced speech recognition technology using the Whisper AI model to convert call audio into text with high accuracy across multiple Indian languages. Once transcribed, the system analyzes the textual data to identify patterns and indicators commonly associated with fraudulent communication, such as requests involving OTPs, KYC updates, bank account details, or urgent UPI transactions. Rather than relying solely on simple keyword detection, the system can be extended to incorporate intelligent natural language processing techniques to better understand context and intent, improving its ability to distinguish between legitimate and malicious interactions. Based on these analyses, the system classifies calls and messages as safe or potentially fraudulent and generates real-time alerts to warn users before they take harmful actions. This multilingual and scalable approach makes the solution particularly effective for India's diverse linguistic landscape, ensuring accessibility and protection for a broad user base while contributing to enhanced security in digital financial transactions.

Problem And Motivation

India’s digital payment ecosystem, particularly UPI, has revolutionized financial transactions by making them fast and accessible. However, this growth has also led to an increase in financial fraud cases, especially through phone calls and SMS scams.Fraudsters often impersonate bank officials or service providers to trick users into sharing sensitive information such as OTPs or account details. These scams are conducted in multiple regional languages, making detection more complex.

Existing solutions are limited as they mainly focus on text-based analysis and English language processing. They fail to address voice-based scams and multilingual communication.The motivation behind this project is to develop a system that can analyze spoken language in real-time and detect fraud across multiple languages, thus enhancing user security and trust in digital payment systems.

Methodology

The system follows a structured pipeline to detect scam calls:

1. Audio Capture: Incoming calls are recorded for analysis.

2. Speech-to-Text Conversion: The Whisper AI model converts audio into text, supporting multiple languages.

3. Keyword Detection: The system scans for keywords such as 'OTP', 'KYC', 'UPI', and 'bank account'.

4. Classification: Calls are labeled as 'Fraud' or 'Safe' based on keyword presence.

5. Output: The result is displayed to the user in real-time.

The implementation is done using Python, integrating audio processing libraries with the Whisper AI model for accurate transcription. The system is designed to work efficiently even with varied accents and noisy audio conditions.

Results

The system was tested with simulated scam and normal calls across different languages.

Observations:

- High transcription accuracy using Whisper AI.

- Effective detection of scam calls containing predefined keywords.

- Ability to handle multilingual inputs.

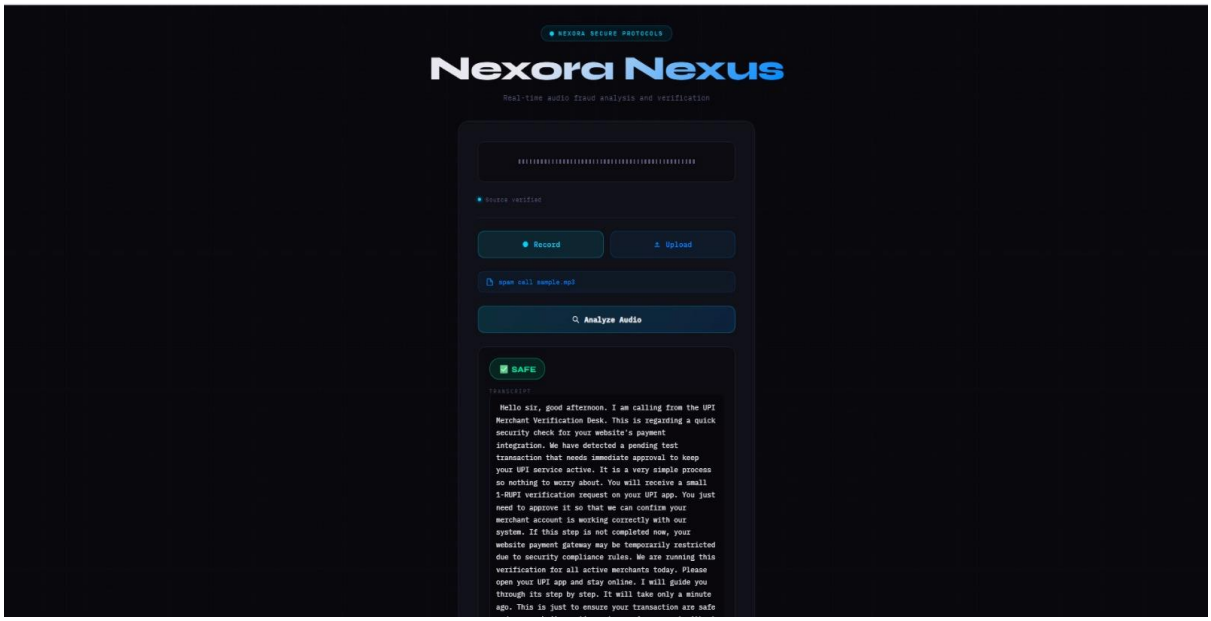
- Near real-time processing capability.

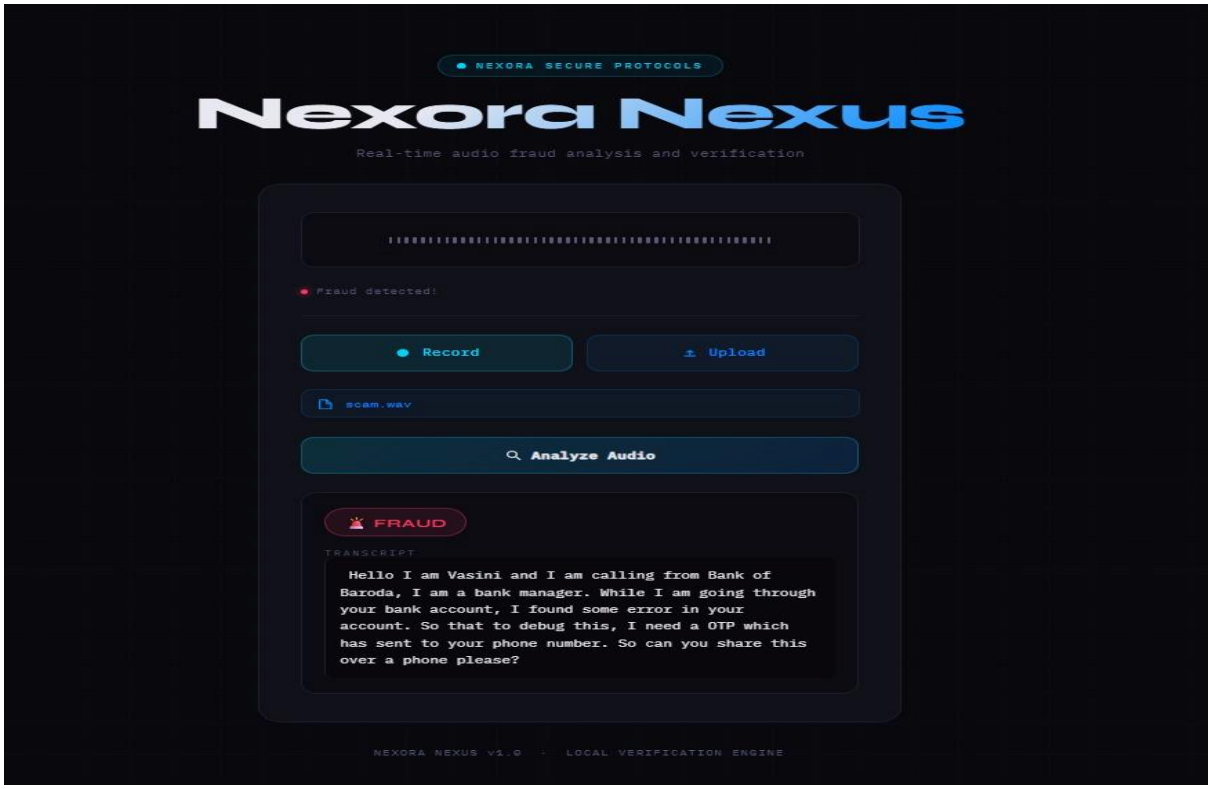
Limitations:

- Occasional false positives when keywords appear in legitimate conversations.

- Reduced accuracy with extremely noisy audio.

Overall, the system demonstrates strong potential in identifying scam calls and improving user awareness.





Conclusion

This project successfully demonstrates a multilingual approach to detecting financial fraud in India's UPI and SMS ecosystem using Whisper AI.By combining speech recognition and keyword-based analysis, the system effectively identifies scam calls in real-time.

While improvements such as context-aware analysis and advanced machine learning classifiers can enhance accuracy, the current system provides a strong foundation for future development.This solution contributes to improving digital security and can be integrated into mobile applications for widespread use.